Ethiopian Association of Anesthetists

Enhanced Recovery after Surgery (ERAS)

Participants' Manual

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Finally, we thank all those who supported this initiative, directly or indirectly, in advancing quality and safety of surgical and anesthesia care provided to breast cancer surgical patients through effective implementation of ERAS protocol. Your commitment to excellence has been instrumental in bringing this important resource to execution.

Introduction

Enhanced Recovery after Surgery (ERAS) programs are patient-centered, evidence-based, multidisciplinary team approaches for perioperative care to reduce the patient's surgical stress response, optimize their physiologic function, and facilitate their recovery time. This care pathway is organized in continuity starting from home, passing through different phases of prehospital/preadmission, preoperative, intraoperative, postoperative period, and back to home again. The ERAS protocols/bundles are multimodal perioperative care pathways which designed to achieve early recovery after surgical procedures by maintaining the baseline preoperative organ function (1).

Enhanced recovery after surgery (ERAS) is a paradigm shift in perioperative care, resulting in substantial improvements in clinical outcomes and cost savings. The implementation of ERAS programs involves a team commonly consisting of surgeons, anesthetists, nurses, clinical pharmacists, nutritionists, physiotherapists, ERAS coordinators and other staff from units that care for the surgical patient (2).

The ERAS Society and others promote, develop, and implement the ERAS programs, publish updated guidelines/evidence that brought dramatic changes to the perioperative care. Some of these prominent changes are promoting carbohydrate drinks about 2 hours prior to surgery than keeping the patient fast for the whole night, minimally invasive surgical approaches instead of large incisions, fluid balance rather than large volumes of intravenous fluid challenges, avoidance/early removal of catheters and drains, early mobilization, and serving of drinks and food as early as possible (3, 4).

The ERAS programs help patients to gain their functional recovery quicker while reducing hospital stays and surgical complications. The implementation of ERAS programs was found reducing perioperative complications, length of hospital stays, and readmissions; all of which have extremely significant cost implications. Despite significant amounts of evidence shows that the ERAS programs lead to improved outcomes, the implementation is too low particularly in the low- and middle-income countries due to multifactorial reasons (5).

Rationale for the CPD Program

The implementation of the ERAS programs was evaluated by several recent studies and resulted in reduction of complication rates, length of hospital stays, costs of perioperative care, while improving patients' overall satisfaction and outcomes in varieties of surgical procedures (4, 6-12). All of these have significant implications for the health systems especially in the low-and middle-income countries such as Ethiopia. The ERAS programs are considered as the current surgical revolutions as they facilitate ready access for perioperative care by accelerating patient turnover, appropriate utilization of expensive medical resources, reduction of morbidity and mortality, and alleviating the economic burdens of healthcare. Therefore, its mandatory to implement the ERAS programs in our country and the implementation needs well-trained multidisciplinary healthcare professionals. In-spite of its benefits, ERAS programs are not being overtly implemented in our country. One of the possible barriers could be absence of well-trained healthcare professionals on the ERAS programs. The ERAS programs are new approaches with innovative paradigm shift and have not been included in pre-service training curriculum of most of healthcare professions yet. Therefore; it is essential to provide in-service training for healthcare professionals who are involved in perioperative care to ensure effective implementation of the ERAS programs. This course on ERAS programs will enable healthcare professionals to implement in their healthcare institutions effectively.

Module Syllabus

1. **Module name**: Enhanced Recovery After Surgery
2. **Module code**: ERASBC 122
3. **Module duration**: 05 days
4. **Pre-requisite**: Being healthcare professional involving in surgical care such as Surgeons, Anesthetists, Nurses, Clinical Pharmacists, Nutritionists, Physiotherapists, Operation Room Heads, Hospital and Healthcare Administrators
5. **Module continuing education unit (CEU) – 15CEU**
6. **Module description**: This 05-days training module/course is designed for healthcare professionals who are involved in surgical care to equip with all the essential knowledge, skills and attitudes on **Enhanced Recovery After Surgery** that enable trainees to provide surgical care targeting faster functional recovery for better patient outcome.
7. **Module/course goal**: Enable healthcare professionals to effectively apply the ERAS bundles of care.
8. **Enabling/learning objectives**: After completion of this course, the trainees will be able to:
   - ✓ Discuss the ERAS bundles of care
   - ✓ Practice ERAS bundles of care
   - ✓ Develop effective leadership role in ERAS
   - ✓ Identify ERAS quality assurance during implementation
9. **Chapters/sessions outlines**:

Chapter	Title /topic	Duration (Minutes)	Sessions
1.	Basic concepts of ERAS	60	1
2.	Principles and roles of ERAS	90	2
3.	Implementation and management of ERAS	250	5
4.	The ERAS bundles of care	600	12

10. Teaching learning methods and activities:
11. Learning materials and resource (including references):
12. **Learning Assessment methods**:
13. Trainer and participants selection criteria:
14. Trainees and participants selection criteria:

15. Training evaluation methods (satisfaction, learning, behavior and impact:
16. Venue: CPD center
17. Composition: Surgical Workforce and Clinical Oncologist

18. ERAS Training Schedule (Day case, minimally invasive surgery)

Time	Day 01	Day 02	Day 03	Day 04
8:30-3:30	Welcome Registration Expectations/prior experience	Preoperative Patient Education and optimization	Thromboembolism prevention	Improving Multidisciplinary communication and education
3:30-4:30	Basic concepts of ERAS (What? Why? How? When?)	Nutrition in ERAS	General and regional anesthesia in ERAS	Possible barriers to implement ERAS (Including Panel Session)
Tea Break				
4:50-5:40	Perioperative complications, functional recovery, and patient experience	Pain Management in ERAS and Opioid pandemics	ERAS in the PACU/recovery room Indwelling catheters (Why ERAS concern about them?)	Quality assurance in ERAS Closure
Lunch				
8:00-8:50	ERAS care bundles, and perioperative goals	Fluid management in ERAS	Epidural and caudal blocks	
Tea Break				

9:10-9:50	The roles of multidisciplinary healthcare professionals in ERAS	Postoperative nausea and vomiting (PONV) prevention	Chest blocks (including erector spinae block)	
9:50-10:30	Implementation and management of ERAS programs	Bowel preparation in ERAS	Abdominal wall blocks with (including quadratus lumborum block, transversalis facia block)	
10:30-10:45	Feedback time	Feedback time	Feedback time	

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Chapter One: Basic Concepts, Principles, and Roles of ERAS

Chapter duration: 170 Minutes

Chapter description: This chapter deals with basic concepts, principles, and roles of ERAS.

Chapter objective: At the end of this chapter, participants will be able to

- ✓ Discuss the basic concepts, principles, and roles of ERAS

Enabling Objective: At the end of this chapter, the participants will be able to

- ✓ Describe basic concepts of ERAS
- ✓ Explain the basic principles of ERAS
- ✓ Discuss the important roles of ERAS
- ✓ Recognize the ERAS care bundles
- ✓ Describe the perioperative goals of ERAS

Sessions

Session 1.1. Basic concepts of ERAS (What? Why? How? When?)

Session 1.2. Perioperative complications, functional recovery, and patient experience

Session 1.3. ERAS care bundles and perioperative goals

Session 1.1. Basic concepts of ERAS

(What? Why? How? When?)

Session duration 90 min

Session objectives

- ✓ Describe the basic concepts of ERAS
- ✓ Discuss importance roles of ERAS in perioperative care

Summary of Activity: basic concepts of ERAS

No	Activity	Method of delivery	Time	Slide Number	Resource
1	Introduction	• Small group discussion • Gallery walk • Interactive lecture	30 min		
2	Elements of ERAS in perioperative care	• Question and Answer • Interactive lecture	15 min		LCD Flip Chart Computer markers
2	Basic corner stone of ERAS	• Interactive lecture • Group discussion	15 min		
3	Benefits of the Enhanced Recovery Programs	• Interactive lecture • Individual reflection	20 min		
4	Summary	• Question and answer • Summarize the session	10 min		

Learning Activities: Basic Concepts of ERAS

S.N	Topic	Time	Activity	
1	Introduction	20 min	• Group discussion • Gallery walk • Describe the basic concepts of ERAS	
2	Elements of ERAS in perioperative care	10 min	• Question and answer • Explain elements of and importance of ERAS	
3	Basic corner stone of ERAS	10 min	• Group discussion • Describe cornerstones of ERAS in perioperative care	
4	Benefits of the Enhanced Recovery Programs	15 min	• Individual reflections • How could Enhanced Recovery Programs reduce the coast?	
5	Summary	10 min	• Question and Answer • Finalize, update and feedback to the participants • Summarize the session	

Session 1.2: Perioperative complications, functional recovery and patient experience

Session duration: 70 Minutes

Session objective: At the end of this chapter, participants will be able to

- ✓ Describe common perioperative complications
- ✓ Discuss the patterns of functional recovery of surgical patients
- ✓ Explain patient experience during the perioperative period

Summary of activity: perioperative complications, functional recovery and patient experience

No	Activity	Method of delivery	Time	Slide Number	Resource
1	introduction	• Reflection on prior experience • Interactive lecture	15		
2	Common perioperative complications	• Small group discussion • Presentation • Interactive lecture	15 min		LCD Flip Chart Computer markers
3	Functional recovery of surgical patient	• Question and answer • Interactive lecture	20 min		
3	Patient experience	• Small group discussion • Interactive lecture	15 min		
4	Summary	• Individual reflections	5 min		

Learning activities; perioperative complications, functional recovery and patient experience

S.N	Topic	Time	Activity
1	Introduction	10 min	• Activity: Question and Answer • Ask the questions to the whole group on previous experiences on complications, functional recovery and patient experience
2	Common perioperative complications	10 min	• Define common perioperative complication • Small group discussion on ERAS role in reducing complication
3	Functional recovery of surgical patient	10 min	• Question and answer • Define functional recovery of surgical patients
4	Patient experience	10	• Group discussion on patient experience • Define patient experience • Group discussion on the difference between traditional and ERAS care patient experience
5	Summary	10	• Individual reflections

Session 1.3. ERAS care bundles and perioperative goals

Session duration: 60 Minutes

Session objective: At the end of this session, participants will be able to;

- ✓ Describe ERAS bundles of care
- ✓ Recognize perioperative goals of ERAS

Summary of Activity: ERAS care bundles and perioperative goals

No	Activity	Method of delivery	Time	Slide Number	Resource
1	Introduction	• Small group discussion • Individual reflection • Interactive lecture	20 min		LCD Flip Chart Computer markers
2	ERAS care bundles	• Small group discussion • Individual reflection • Interactive lecture	15 min		
3	Perioperative goals of ERAS	• Small group discussion • Individual reflection • Interactive lecture	15 min		
4	Summary	• Individual reflection	10 min		

		• Session summary			

Learning activities: ERAS care bundles and perioperative goals

No	Topic	Time	Activity
1	Introduction	15 min	• Small group discussion • Individual reflection • Describe the ERAS bundles of care?
2	ERAS care bundles	10 min	• Small group discussion • Individual reflection • What are the ERAS care bundles
3	Perioperative goals of ERAS	10 min	• Small group discussion • Individual reflection • What are perioperative goals of ERAS
4	Summary	10 min	• Individual reflection • Session summary

Chapter Two: The Implementation and Management of ERAS programs

Chapter duration: 150 Minutes

Chapter description: This chapter deals with the implementation and management of ERAS programs

Chapter objective: At the end of this chapter, participants will be able to

- ✓ Discuss the appropriate modalities and barriers of implementing and managing ERAS programs

Enabling Objective: At the end of this chapter, the participants will be able to

- ✓ Describe the roles of multidisciplinary healthcare professionals in ERAS
- ✓ Discuss implementation approaches of ERAS programs
- ✓ Elaborate management principles of ERAS programs
- ✓ Develop effective multidisciplinary communication and education in ERAS teams
- ✓ Discuss possible barriers to the implementation of the ERAS
- ✓ Recognize quality assurance modalities in ERAS

Sessions

Session 2.1. The roles of multidisciplinary healthcare professionals in ERAS

Session 2.2. Implementation and management of ERAS programs

Session 2.3. Improving Multidisciplinary communication and education

Session 2.4. Possible barriers to the implementation of the ERAS programs

Session 2.5. Quality assurance in ERAS

Session 2.1. The roles of multidisciplinary healthcare professionals in ERAS

Session duration: 50 Minutes

Session Objectives: At the end of this session participants will be able to:

- ✓ List roles of surgeons in ERAS
- ✓ Identify roles of anesthetists in ERAS protocol
- ✓ List roles of Nurses in ERAS
- ✓ Identify roles of pharmacists in ERAS
- ✓ List roles of dieticians in ERAS

Summary of Activity: The roles of multidisciplinary healthcare professionals in ERAS

No	Activity	Method of delivery	Time	Slide Number	Resource
1	Introduction	• Group discussion • Interactive lecture	10 min		
1	The roles of multidisciplinary healthcare professionals in ERAS	• Role play • Interactive lecture	30 Min		
3	Summary	• Question and Answer	10 min		

Learning activities; the roles of multidisciplinary healthcare professionals in ERAS

S.N	Topic	Time	Activity
1	Introduction	5	• Group discussion • Describe the importance of mentioning the roles of health professionals in ERAS

2	The roles of multidisciplinary healthcare professionals in ERAS	20 Minutes	• Role play • Show the roles of health professionals involved in ERAS (surgeon, Anesthetist, nurse, pharmacist, dietitians, physiotherapist)
3	Summary	10	• Question and answer • Summarize the session

Session 2.2.; Implementation and management of ERAS programs

Session duration: 90 Minutes

Session objectives: At the end of this session participants will be able to:
✓ Discuss the steps of implementation process of ERAS

Summary of Activity: Implementation and management of ERAS programs

No	Activity	Method of delivery	Time	Slide Number	Resource
1	Implementation process of ERAS programs	• Interactive lecture • Gallery walk • Question and answer	50 minute		
2	Management of ERAS	• Group discussion • Interactive lecture	30		
3	Summary	• Individual reflection • Summarize the session	10		

Learning activities; Implementation and management of ERAS programs

S.N	Topic	Time	Activity
1	Implementation process of ERAS programs	35 minute	• Be in group and tell the participants to develop a presentation for Gallery walk (the participants should be in 8 group, each will be provided one implementation process step to write and present it within allocated time

			• Explain every steps of the implementation process as summary form • Ask participants on steps and give time to answer • Summarize the session
	Management of ERAS	20	• Group discussion
	Summary	10	• Individual reflection

Session 2.3. Possible Barriers of Implementing ERAS

Session Duration; 60 min

Session objectives: At the end of this session participants will be able to:
✓ Explain possible barriers to implement ERAS programs

Summary of Activity: Possible barriers to the implementation of the ERAS programs

No	Activity	Method of delivery	Time	Slide Number	Resource
1	Introduction	• *Small group discussion and gallery walk* • *Interactive lecture*	20 min		
2	Possible barriers to the implementation of the ERAS programs	• Interactive lecture • Question and answer	35 min		
3	Summary	• Individual reflection	5 min		

Learning activities; Common barriers to the implementation of the ERAS programs

S.N	Topic	Time	Activity
1	Introduction	15 min	• *Small group discussion and gallery walk* • Describe ERAS implementation barriers
2	common barriers to the implementation of the ERAS programs	25	• Form group to discuss barriers of ERAS implementation • Provide time to present their answer • Describe every barriers in implementing ERAS • Ask participants to reflect each barriers

			• Summarize the session •
3	Summary	5	• Individual reflection

Session 2.4. Improving Multidisciplinary Professional Communication in ERAS

Session duration: 60 Minutes

Session Objectives: At the end of this session participants will be able to:

- ✓ Identify ways of effective professional communication
- ✓ Describe Multidisciplinary knowledge transfer
- ✓ Identify ways of improving Multidisciplinary professional communication

Summary of Activity: Improving Multidisciplinary communication and education

No	Activity	Method of delivery	Time	Slide Number	Resource
1	Introduction	• Small group discussion and reflection	15 min		
2	Effective professional communication	• Group discussion • Interactive lecture • Questions and answer	15 min		
3	Multidisciplinary knowledge transfer	• Group discussion • Interactive lecture • Questions and answer	15 min		
4	Improving multidisciplinary professional communication	• Group discussion • Interactive lecture • Questions and answer	15 min		

Learning activities: Improving Multidisciplinary communication and education

S.N	Topic	Time	Activity
1	Introduction	10 minute	• Small group discussion and reflection • Describe multidisciplinary professional communication
2	Effective professional communication		• Group discussion • Question and answer • Describe components of Multidisciplinary professional communication?
3	Multidisciplinary knowledge transfer		• Group discussion • Question and answer • Describe characteristics of effective multidisciplinary professional communication
4	Improving multidisciplinary professional communication		• Group discussion • Question and answer • List advantages of effective multidisciplinary professional communication

Session 2.5. Quality assurance steps in ERAS

Session duration: 60 minutes

Session Objectives: At the end of this session participants will be able to: • Describe quality assurance steps • Recognize tools used in quality assurance of ERAS programs

Summary of Activity: Quality assurance steps in ERAS

No	Activity	Method of delivery	Time	Slide Number	Resource
1	Introduction	Small group discussion and presentation	20 min		
2	Steps of quality assurance in ERAS	• Question and answer • Interactive lecture	10 min		
3	Tools used in quality assurance of ERAS	• Question and answer • Interactive lecture	10 min		

Learning activities; Quality assurance steps in ERAS

S.N	Topic	Time	Activity

1	Introduction	20 minute	• Provide time for participants to form a group and to discuss on two topics (group 1 quality assurance steps group 2 tool used in quality assurance • Provide time to form presentation and discuss on each topic • Define quality assurance in ERAS • Ask question on quality assurance and provide time for participants to answer the questions • Summarize the session
	Steps of quality assurance in ERAS		• Question and answer • List and explain each steps of quality assurance
	Tools used in quality assurance of ERAS		• Question and answer • List and explain tools used in quality assurance

Chapter Three: The ERAS Bundles of Care

Duration: 720minutes

Chapter description: This Chapter focuses on ERAS care bundles that are evidence-based, multidisciplinary protocols that aim to optimize patient outcomes in the perioperative period.

Chapter Objective: At the end of this Chapter participants will be able to:

- Explain the ERAS care bundles recommended during perioperative period aiming for better patient outcome and reducing hospital stay and coast.

Enabling Objectives: At the end of this Chapter participants will be able to

- Define ERAS care bundles
- Identify different ERAS care bundles recommended during preoperative, intraoperative and postoperative period
- Identify ways of preoperative pt education and optimization
- Identify how to provide and manage nutrition in ERAS
- Describe pain protocol and opioid pandemics
- Determine fluid management principles in ERAS
- Explain how to prevent postoperative nausea and Vomiting
- Identify ways of bowel preparation in ERAS protocol
- Elaborate how to prevent thromboembolism during perioperative period
- Explain anesthesia management principles in ERAS
- Identify ERAS principles and implementation in PACU/recovery room
- Identify and

Sessions

Session 3.1 Preoperative Patient Education and optimization

Session 3.2 Nutrition in ERAS

Session 3.3 Pain Management in ERAS and Opioid pandemics

Session 3.4 Fluid management in ERAS

Session 3.5 Postoperative nausea and vomiting (PONV) prevention

Session 3.6 Bowel preparation in ERAS

Session 3.7 Thromboembolism prevention

Session 3.8 General and regional anesthesia management in ERAS

Session 3.9 ERAS in the PACU/recovery room

Session 3.10 Epidural and caudal blocks

Session 3.11 Chest blocks (including erector spinae block)

Session 3.12 Abdominal wall blocks

Session 3.1 Preoperative Patient Education and optimization

Duration 110 Minutes

Session objectives
• Describe components of patient education • Identify concepts of effective communication • List mechanism of Perioperative risk analysis in an Enhanced Recovery after Surgery • Describe concepts of Informed consent • List Multidisciplinary optimization of common comorbidities • List Impacts and management of preoperative anxiety

Summary of Activity: Preoperative Patient Education and optimization

No	Activity	Method of delivery	Time	Slide Number	Resource
1.	Introduction	• Gallery walk • Interactive Lecture	40 min		LCD Flip Chart Computer markers
2.	Effective communication	• Interactive lecture • Group Discussions	10 min		
3.	Perioperative risk analysis in an Enhanced Recovery after Surgery and informed consent	• Interactive lecture • Group Discussions	20 min		
4.	Multidisciplinary optimization of common comorbidities	• Group discussion • Individual reflection	20 min		
5.	Impacts and management of preoperative anxiety	• Interactive lecture • Question and answer	10 min		

6.	Summary	• Question and Answer • Individual reflection	10 min		

Learning activities: Preoperative Patient Education and optimization

	Topic	Time	Activity
1	Introduction	30 min	• Gallery walk of the whole group
2	Effective communication	5min	• Discussion of the whole group on definition of effective communication • Elaborate the characteristics of effective communication
4	Perioperative risk analysis in an Enhanced Recovery after Surgery and informed consent	10 min	• Discuss on perioperative risk analysis • Describe about ways and advantages of informed consent
5	Multidisciplinary optimization of common comorbidities	20 min	• Discuss on multidisciplinary involvement in optimization of patients • Individual reflection about common comorbidities during perioperative period
6	Impacts and management of preoperative anxiety	5 min	• Question and answer on impacts of perioperative anxiety • Describe perioperative anxiety
	Summary	10	• Question and answer • Individual reflection of the whole session

Session 3.2 Nutrition in ERAS

Duration 60 minutes

Session objectives; at the end of this session participants will be able to:
Identify the impact of nutritional status on surgical outcome, preoperative nutritional assessment and nutritional supplement and optimization

Summary of Activity: Nutrition in ERAS

No	Activity	Method of delivery	Time	Slide Number	Resource
1.	Introduction	• Group discussion • Interactive Lecture	10 min		LCD Flip Chart Computer markers
2.	Screen for nutrition risk before surgery	• Interactive lecture • Group Discussions	15 min		
3.	Avoid fasting with oral carbohydrate loading	• Interactive lecture • Group Discussions	15 min		
4.	Begin oral nutrition early postoperatively	• Any questions	15 min		
5.	Summary	• Present the final session • Ask questions and comments.	5 min		

Learning activities: Nutrition in ERAS

	Topic	Time	Activity

1	Introduction	5 min	• Discuss with the whole group
2	Screen for nutrition risk before surgery	10min	• Discuss on screening modalities in risk assessment for nutrition in ERAS • Describe the types of foods recommended in ERAS
3	Avoid fasting with oral carbohydrate loading	10 min	• Discuss on disadvantages of prolonged fasting • Describe Elaborate dietitians involvement for nutrition in ERAS
4	Begin oral nutrition early postoperatively	10	• Discuss on principles of nutritional support in ERAS
5	Summary	5	• Question and answer on recommendations of nutritional support in ERAS

Session 3.3 Pain Management in ERAS and Opioid pandemics

Session duration: 60 Minutes

Session objective: At the end of this chapter, participants will be able to

- ✓ Describe the consequences of untreated acute perioperative pain
- ✓ Use different pain assessment tools
- ✓ Discuss appropriate modalities of pain management
- ✓ Explain opioid pandemics
- ✓ Recognize adverse effects of opioids in ERAS perspectives

Summary of Activity: Pain Management in ERAS and Opioid pandemics

No	Activity	Method of delivery	Time	Slide Number	Resource
1.	Introduction	• Group discussion • Gallery walk • Individual reflection • Interactive Lecture	20 min		LCD Flip Chart Computer markers
2.	Pain assessment	• Group Discussions and presentation • Interactive lecture	40 min		
3.	Pain management modalities in ERAS	• Group discussion and reflection • Interactive lecture	15 min		
4.	Opioid pandemics in ERAS perspectives	• Group discussion and reflection • Interactive lecture	20 min		
5.	Practical points in pain management in ERAS	• Interactive lecture	15 min		
6.	Summary	• Question and answer	5 min		

		• Individual reflection			

Learning activities: Pain Management in ERAS and Opioid pandemics

	Topic	Time	Activity
1	Introduction	13 min	• Discuss with the whole group • Gallery walk • Individual reflection
2	Pain assessment	25 min	• Discuss on different pain assessment methods (including subjective and objective).
3	Pain management modalities in ERAS	10 min	• Group discussion and reflection
4	Opioid pandemics in ERAS perspectives	13 min	• Group discussion and reflection
	Practical points in pain management in ERAS	5 min	• Question and answer
	Summary	5	• Question and answer • Individual reflection

Session 3.4 Fluid management in ERAS

Session duration: 50 Minutes

Session specific objectives:

- Identify Perioperative hydration assessment and monitoring
- Describe goal directed perioperative fluid therapy

Summary of Activity: Fluid management in ERAS

No	Activity	Method of delivery	Time	Slide Number	Resource
7.	Introduction	• Group discussion • Interactive Lecture	20 min		LCD Flip Chart Computer markers
8.	Perioperative hydration assessment and monitoring	• Interactive lecture • Group Discussions	15 min		
9.	Goal directed perioperative fluid therapy	• Interactive lecture • Group Discussions	15 min		
10.	Questions	• Any questions	5 min		
11.	Summary	• Present the final session • Ask questions and comments.	5 min		

Learning activities: Fluid management in ERAS

	Topic	Time	Activity
1	Introduction	10 min	• Discuss with the whole group

2	Perioperative hydration assessment and monitoring	10min	• Discuss on hemodynamic parameters monitoring during fluid administration • Describe Monitoring standards recommended for this
4	Goal directed perioperative fluid therapy	10 min	• Discuss on goal directed fluid therapy • Describe the advantages of goal directed fluid therapy

Session 3.5 Postoperative nausea and vomiting (PONV) prevention

Session duration:40 minutes

Session objectives at the end of this session
Define basic pathophysiology of PONV Improve Impacts of PONV Assess the risk of having PONV Increase Prevention and management of PONV

Summary of Activity: Postoperative nausea and vomiting (PONV) prevention

No	Activity	Method of delivery	Time	Slide Number	Resource
1.	Introduction	• Individual reflections • Interactive lecture	10 min		LCD Flip Chart Computer markers
2.	The risks and impacts of having PONV	• *Group discussion and gallery walk*	15 min		
3.	Prevention of PONV	• Interactive lecture • Question and answer	10 min		
4.	Summary	• Summarize the session • Question and answer.	5 min		

Learning activities: Postoperative nausea and vomiting (PONV) prevention

	Topic	Time	Activity

1	Introduction	5 min	• Individual reflections
2	The risks and impacts of having PONV	10 min	• *Group discussion and gallery walk*
3	Prevention of PONV	5 min	• Question and answer •
4	Summary	5 min	• Question and answer

Session 3.6: Bowel preparation in ERAS

Session duration 60 minutes

Session objectives at the end of this session participants will be able to
Provide appropriate bowel preparation
Implement ERAS recommendations during bowel preparation

Session 3.7 Thromboembolism prevention

Session duration: 60 Minutes

Session objective: At the end of this chapter, participants will be able to

- ✓ Describe risk factors for thromboembolism
- ✓ Explain assessments of the risks of thromboembolism
- ✓ Discuss thromboembolism prevention in ERAS perspective

Summary of Activity: Thromboembolism prevention

No	Activity	Method of delivery	Time	Slide Number	Resource
5.	Introduction	• Individual reflections • Interactive Lecture	15 min		LCD Flip Chart Computer markers
6.	Risk factors for thromboembolism and assessment	• Interactive lecture • Group discussion • Gallery walk	20 min		
7.	Prevention of thromboembolism	• Interactive lecture • Case presentation	20 min		
8.	Questions	• Any questions	5 min		
9.	Summary	• Summarize the session • Individual reflection.	5 min		

Learning activities: Thromboembolism prevention

	Topic	Time	Activity

1	Introduction	10 min	• Discuss with the whole group
2	Risk factors for thromboembolism and assessment	10 min	• Discuss with the whole group on risk assessment methods • Describe the risks of thromboembolic disorders
3	Prevention of thromboembolism	10 min	• Case presentation on thromboembolic disorders •
4	Summary	5 min	• Individual reflection of session summary

Session 3.8: General and regional anesthesia management in ERAS

Duration: 60 Minutes

Session Objectives
➢ Identify General anesthesia management principles in ERAS ➢ Describe regional Anesthesia management in ERAS

Summary of Activity: General and regional anesthesia management in ERAS

No	Activity	Method of delivery	Time	Slide Number	Resource
10	Introduction	• Group discussion • Interactive Lecture	15 min		LCD Flip Chart Computer markers
11	General anesthesia management principles in ERAS	• Interactive lecture • Individual reflection	20 min		
12	Regional Anesthesia management in ERAS	• Interactive lecture • Individual reflection	15 min		
13	Questions	• Any questions	5 min		
14	Summary	• Summarize the session • Individual reflection.	5 min		

Learning activities: General and regional anesthesia management in ERAS

	Topic	Time	Activity
1	Introduction	10 min	• Discuss with the whole group

2	General anesthesia management principles in ERAS	12 min	• 4-5 trainees reflect individually • Describe General anesthesia management principles in ERAS
3	Regional Anesthesia management in ERAS	10 min	• 4-5 trainees reflect individually • Describe regional anesthesia management principles in ERAS
4	Summary	5 min	• Individual reflection of session summary

Session 3.9: ERAS in the Post-Anesthetic Care Unit (PACU)

Session duration: 60 Minutes

Session objective: At the end of this session, participants will be able to; ✓ Describe care protocols in the PACU in ERAS perspectives

Summary of Activity: ERAS in the Post-Anesthetic Care Unit (PACU)

No	Activity	Method of delivery	Time	Slide Number	Resource
1	Introduction	• Group discussion and reflection • Interactive Lecture	15 min		LCD Flip Chart Computer markers
2	Main activities in PACU	• *Small group discussion* • *Presentation* • *Interactive lecture*	40 min		
3	Summary	• Question and answer • Individual reflection	5 min		

Learning activities: ERAS in the Post-Anesthetic Care Unit (PACU)

	Topic	Time	Activity
1	Introduction	10 min	• Presentation Group discussion and reflection • How could ERAS be applied in PACU
2	Main activities in PACU	20 min	• *Small group discussion* • *Presentation*

			• *Describe main activities in ERAS(discharge criteria, pain management, nausea and vomiting management)*
3	Summary	5 min	• Question and Answer • Individual reflection • Summarize the session

Session 3.10: Epidural and caudal blocks

Duration 50 minutes

Session objectives
Perform epidural and caudal blocks in accordance with ERAS recommendations

Summary of Activity: Epidural and caudal blocks

No	Activity	Method of delivery	Time	Slide Number	Resource
1	Introduction	• Individual reflection • Interactive lecture	15 min		LCD Flip Chart Computer markers
2	Techniques of epidural block	• Group discussion and reflection • Interactive lecture	20 min		
3	Techniques of caudal block	• Interactive lecture • Question and answer	20 min		
4	Assessing block and troubleshooting	• group discussion and reflection • Interactive lecture	15 min		

Learning activities: Epidural and caudal blocks

	Topic	Time	Activity
1	Introduction	10 min	• Individual reflection • Describe the importance of epidural and caudal blocks as ERAS protocol.

2	Techniques of epidural block	10 min	• Group discussion and reflection • Describe the structures the needle goes through until it reach epidural space.
3	Techniques of caudal block	10 min	• Question and Answer
4	Assessing block and troubleshooting	10 min	• Group discussion and reflection • Describe how to assess the level of the block and guarantee surgical anesthesia

Session 3.11: Chest wall blocks in ERAS

Session duration: 60 Minutes

Session objective: At the end of this session, participants will be able to;

- ✓ Describe common chest wall blocks in ERAS
- ✓ Recognize indications of chest wall blocks
- ✓ Discuss the complications of chest wall blocks

Summary of Activity: Chest wall blocks in ERAS

No	Activity	Method of delivery	Time	Slide Number	Resource
1	Introduction	• Group discussion • Presentation • Interactive Lecture	25 min		LCD Flip Chart Computer markers
2	Paravertebral block	• Small group discussion • gallery walk • interactive lecture	12 min		
3	Intercoastal/subcoastal blocks (ICIS)	• Small group discussion • gallery walk • interactive lecture	12 min		
4	Erector Spinae block (ESB)	• Small group discussion • gallery walk • interactive lecture	11 min		

Learning activities: Chest wall blocks in ERAS

	Topic	Time	Activity
1	Introduction	15 min	• Group discussion • Presentation • Describe commonly used chest blocks and benefits
2	Paravertebral block	7min	• Small group discussion • gallery walk • Describe complications of paravertebral block
3	Intercoastal/subcoastal blocks (ICIS)	7 min	• Small group discussion • gallery walk • Describe complications of Intercoastal/subcoastal block
4	Erector Spinae block (ESB)	6 min	• Small group discussion • gallery walk • Describe complications of Erector Spinae block

Session 3.12: Abdominal field blocks in ERAS

Session duration: 60 Minutes

Session objective: At the end of this session, participants will be able to;

- ✓ Describe common abdominal field blocks in ERAS
- ✓ Recognize indications of abdominal field blocks
- ✓ Discuss the complications of abdominal field blocks

Summary of Activity: Abdominal field blocks in ERAS

No	Activity	Method of delivery	Time	Slide Number	Resource
30	Introduction	• Group discussion and reflection • Presentation • Interactive Lecture	40 min		LCD Flip Chart Computer markers
31	Trans-abdominis plane block (TAP)	• Interactive lecture • Question and answer	10 min		
32	Ilio-inguinal-ilio-hypogastric block (ILIG)	• Interactive lecture • Question and answer	10 min		
33	Quadratus Lumborum Block (QLB)	• Interactive lecture • Question and answer	10 min		
34	Rectus Sheath Block	• Interactive lecture • Question and answer	10 min		
35	Assessment and monitoring of abdominal field blocks	• Interactive lecture • Discussion and Individual reflection	10 min		

36	Summary	• Interactive lecture • Group discussion and reflection • Ask questions and comments.	20 min		

Learning activities: Abdominal field blocks

	Topic	Time	Activity
1	Introduction	30 min	• Discuss with the whole group • Presentation
2	Trans-abdominis plane block (TAP)	5 min	• Question and Answer
3	Ilio-inguinal-ilio-hypogastric block (ILIG)	5 min	Question and Answer
4	Quadratus Lumborum Block (QLB)	5 min	Question and Answer
	Rectus Sheath Block	5 min	Question and Answer
	Assessment and monitoring of abdominal field blocks	5 min	• Discussion and individual reflection
	Summary	15 min	• Group discussion and reflection • Ask questions and comments

Annex 1: Think-Pair Share Template

Name _____ Date_____

Think-Pair-Share

Read the following question or problem:

Think

On your own, write three ideas you have about this question or problem:

1._____

2._____

3._____

Pair

Discuss your ideas with a partner. Put a check by any ideas, above, that your partner also wrote down. Then, write down ideas your partner had that you did not have:

1._____

2._____

3._____

Share

Review all of your ideas and circle the one you think is most important. One of you will share this idea with the whole group. As you listen to the ideas of the whole group, write down three more ideas you liked:

1._____

2._____

3._____

Annex 2: Gallery Walk

A gallery walk is a great way to engage participants in a training session. It can be used to introduce a new topic, review material, or even to assess understanding. Here are some instructions for conducting a successful gallery walk:

1. Prepare the space: Before the gallery walk, make sure the space is set up for success. Arrange the room so that participants can easily move around and view the materials. If possible, provide tables or stands to display the materials.

2. Prepare the materials: Gather the materials that you want to use for the gallery walk. These can include posters, handouts, pictures, videos, or any other visual aids. Make sure the materials are clearly labeled and organized in a way that makes sense.

3. Introduce the activity: Explain to the participants what a gallery walk is and how it will be conducted. Let them know that they will be moving around the room to view the materials and that they should take notes or ask questions as they go.

4. Divide into groups: Divide the participants into small groups and assign each group to a specific area of the room. Make sure each group has enough time to view the materials and discuss them.

5. Allow time for discussion: After each group has had a chance to view the materials, allow time for discussion. Encourage participants to ask questions and share their thoughts and ideas.

6. Wrap up: At the end of the gallery walk, review the materials and discuss any key points or takeaways. Make sure to thank the participants for their participation and encourage them to ask any remaining questions.

By following these instructions, you can ensure that your gallery walk is a successful and engaging activity for your training session. It can be a great way to introduce a new topic, review material, or assess understanding

Annex 3: Instructional Content Analysis

Title of the training: **Enhanced Recovery After Surgery (ERAS)**

Job/service to improved /enhanced: Perioperative Care Service

Target audience: Surgeons, Anesthetists, Nurses, Clinical Pharmacists, Nutritionists, Physiotherapists, Operation Room Heads, Hospital and Healthcare Administrators

Major Task	Knowledge Analysis	Skill and Attitude Analysis
Patient Education and Multidisciplinary Optimization	➢ Educate patients on to dos, not to dos, and adherence ➢ Effective communication ➢ Patient rights ➢ Perioperative risk analysis ➢ Informed consent ➢ Multidisciplinary optimization of common comorbidities ➢ Impacts and management of preoperative anxiety	➢ Deliver appropriate preoperative patient education ➢ Effective communication skills ➢ Receive patient feedback ➢ Obtain informed consent ➢ Provide multidisciplinary preoperative optimization ➢ Manage preoperative anxiety
Nutritional Assessment and Optimization	➢ The impact of nutritional status on surgical outcome ➢ Preoperative nutritional assessment ➢ Nutritional supplement and optimization	➢ Assess nutritional status ➢ Provide nutritional supplement and optimization
Pain Management	➢ Basic physiology of acute pain ➢ Impacts of uncontrolled acute pain	➢ Assess acute pain ➢ Provide optimal acute pain management

	➢ Acute pain assessment ➢ Multimodal analgesia ➢ Preemptive analgesia	➢ Provide multimodal analgesia ➢ Administer preemptive analgesia
Fluid Management	➢ Perioperative hydration assessment and monitoring ➢ Goal directed perioperative fluid therapy	➢ Assess and monitor hydration status ➢ Provide goal directed fluid therapy
Postoperative Nausea and Vomiting (PONV) Prevention	➢ Basic pathophysiology of PONV ➢ Impacts of PONV ➢ Assess the risk of having PONV ➢ Prevention and management of PONV	➢ Assess the risk of having PONV ➢ Prevent PONV ➢ Manage PONV
Manage Bowel Preparation	➢ Impacts of bowel preparation ➢ Techniques of bowel preparation ➢ ERAS recommendations on bowel preparation	➢ Provide appropriate bowel preparation ➢ Implement ERAS recommendations during bowel preparation
Thromboembolism Prevention	➢ Assess the risk of the occurrence of thromboembolic events ➢ Prevention of thromboembolic events ➢ Management of thromboembolic events	➢ Assess the risks for thromboembolic events ➢ Prevent thromboembolic events ➢ Manage thromboembolic events
Regional Anesthesia in ERAS	➢ Spinal and epidural anesthesia in ERAS ➢ Chest wall blocks (including erectus spinae block)	➢ Perform neuraxial and field blocks in accordance with ERAS recommendations

	➢ Abdominal wall blocks (including quadratus lumborum block and transversalis fascia block)	
Manage and Reverse Neuromuscular Blockade	➢ Impacts of neuromuscular blockade on recovery ➢ Reversal of neuromuscular blockade	➢ Appropriate administration and monitoring of neuromuscular blockade ➢ Appropriate reversal of neuromuscular blockade
Mange Indwelling Catheters and Drains	➢ Impacts of indwelling catheters and drains ➢ ERAS recommendations on the use of catheters and drains	➢ Appropriate use of catheters and drains according to ERAS recommendations
Use of Opioids	➢ Opioid pandemics ➢ Side-effects of opioids ➢ Opioid sparing strategies ➢ Appropriate utilization of opioids	➢ Reduce side-effects of opioids ➢ Apply opioid-sparing strategies ➢ Safe utilization of opioids
Audit and Predictive Analysis	➢ Audit in ERAS ➢ Predictive analysis	➢ Conduct audit in ERAS ➢ Perform predictive analysis
Multidisciplinary Communication and Education	➢ Effective professional communication ➢ Multidisciplinary knowledge transfer	➢ Perform effective professional communication ➢ Facilitate multidisciplinary knowledge transfer
Implementation of ERAS Programs	➢ Leadership in ERAS programs	➢ Implement ERAS programs